

Cambridge (CIE) IGCSE Physics
Paper 4: Theory (Set B)
Extended

Saturday 26 April 2025

Morning (Time: 1 hour 15 minutes)

Total marks

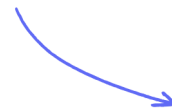
/ 80

Instructions

- Try to complete this mock exam paper in one sitting, under exam conditions. Use all the time available and check your answers to each question at the end before submitting.
- Remember this is PRACTICE. Mistakes are fine and will help you improve in time for the real exam - just do your best.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8N (acceleration of free fall = 9.8m/s^2).
- You should show all your working and use appropriate units.

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or visit the mock exams landing page for this course



- 1 (a)** A rocket is stationary on the launchpad. At time $t = 0$, the rocket engines are switched on, and exhaust gases are ejected from the nozzles of the engines. The rocket accelerates upward.

Fig. 1.1 shows how the acceleration of the rocket varies between time $t = 0$ and time $t = t_f$.

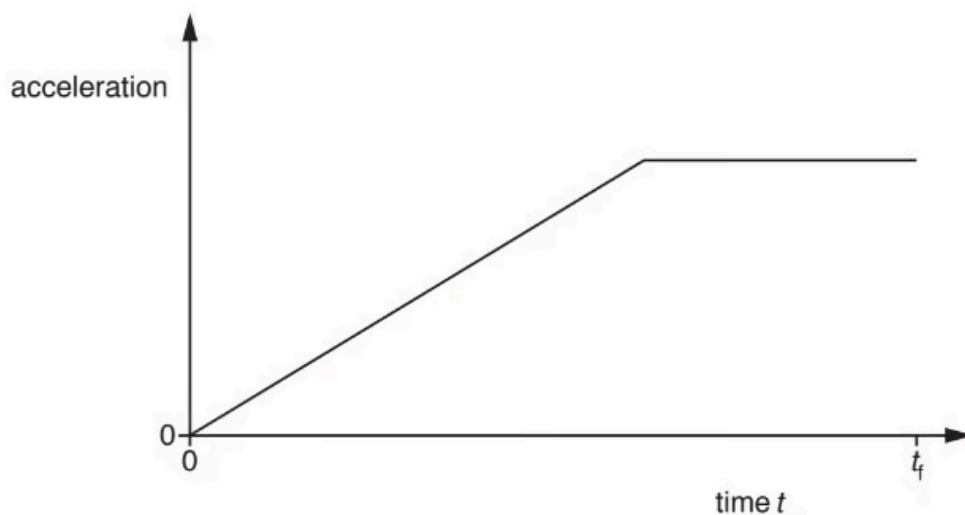


Fig. 1.1

Define *acceleration*.

(1 mark)

- (b)** On Fig. 1.2, sketch a graph to show how the speed of the rocket varies between time $t = 0$

and time $t = t_f$.



Fig. 1.2

(3 marks)

- (c)** A rocket is now far from the Earth. The effect of the Earth's gravity on the motion of the rocket is insignificant. As the rocket accelerates, its momentum increases.

(i) State the principle of the conservation of momentum.

[2]

(ii) Explain how the principle of the conservation of momentum applies to the accelerating rocket and the exhaust gases.

[2]

.....

.....

.....

(4 marks)

2 (a) Fig. 2.1 shows a train.

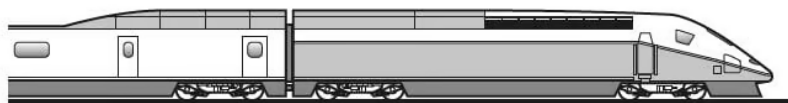


Fig. 2.1

The total mass of the train and its passengers is 750 000 kg. The train is travelling at a speed of 84 m/s. The driver applies the brakes and the train takes 80 s to slow down to a speed of 42 m/s.

Calculate the impulse applied to the train as it slows down.

impulse =

.....

.....

.....

(3 marks)

(b) Calculate the average resultant force applied to the train as it slows down.

force =

.....

.....

(2 marks)

(c) Suggest how the shape of a modern electric train helps it to travel at high speeds.

.....

(1 mark)

- (d)** The train took 80 s to reduce its speed from 84 m/s to 42 m/s. Explain why, with the same braking force, the train takes less than 80 s to reduce its speed from 42 m/s to zero.

(1 mark)

- (e)** On a wet day, the train travels a greater distance before it stops along the same track. The train has the same speed of 84 m / s before the brakes are applied.

Suggest a reason for this.

(1 mark)

- 3 (a)** A rifle fires a bullet of mass 0.020 kg vertically upwards through the air. As it leaves the rifle, the speed of the bullet is 350 m/s.

Calculate

- (i) the energy in the kinetic store of the bullet as it leaves the rifle,

energy =[3]

- (ii) the maximum possible height that the bullet can reach.

maximum height =[2]

.....

.....

.....

.....

.....

(5 marks)

(b) The actual height reached by the bullet is less than the value calculated in **(a)(ii)**.

(i) Explain, in terms of the forces acting on the bullet, why this is so.

[2]

(ii) As the bullet rises through the air, the amount of energy in its kinetic store decreases.

State what happens to this energy.

[2]

(4 marks)

4 (a) Describe the movement of the molecules in

(i) a solid.

[1]

(ii) a gas.

[2]

.....
.....
.....
(3 marks)

(b) A closed box contains gas molecules. How does the momentum of the molecules cause them to exert a pressure on the walls of the box?

.....
.....
.....
.....
(4 marks)

- 5 (a)** Complete the sentences with words that describe the main process of energy transfer by heating in each case.

A man goes for a walk on a cold day. He touches a metal gate, which removes energy from his hands by He holds the sides of a cup containing a hot drink. His hands gain energy by Some farm workers have lit a fire. The man warms his hands by the side of the fire. His hands gain energy by

(3 marks)

- (b)** Describe in terms of particles the transfer of energy from the thermal store of the man's hands through the metal of the gate.

(2 marks)

- (c)** Fig. 5.1 shows a car on a sunny day in a hot country.

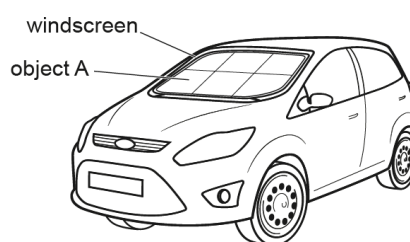


Fig. 5.1

The object labelled A is placed inside the windscreen. It is used by the owner of the car to reduce the temperature rise of the air in the car.

Select the most suitable material for the outer surface of object A. Explain your choice.

dull black	dull white	shiny black	shiny white
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(2 marks)

- 6 (a)** Fig. 6.1 shows an arrangement of glass prisms inside a box. The angles of the prisms are 45° , 45° and 90° .

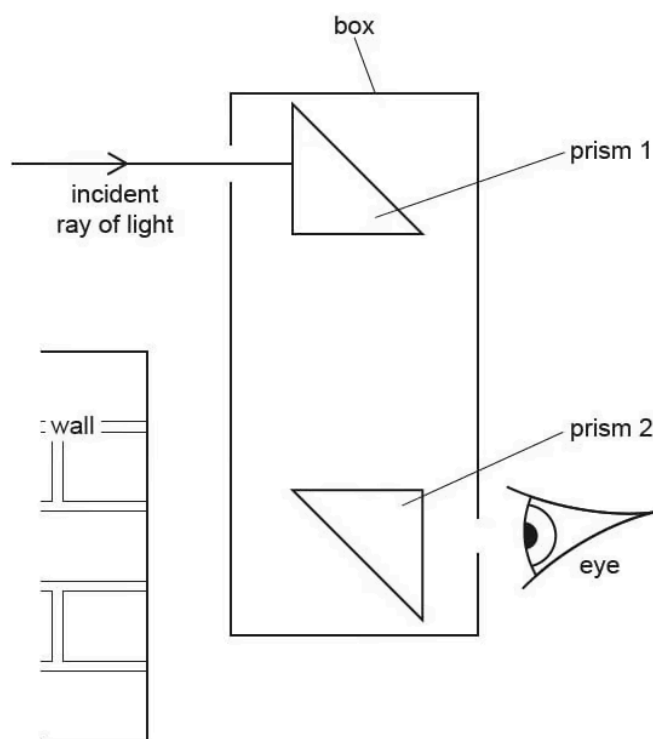


Fig. 6.1 (not to scale)

This is a device used to view objects that are behind a wall. The incident ray of light undergoes total internal reflection in the prisms.

On Fig. 6.1, complete the path of the ray through the device and show the ray as it emerges from the box.

(3 marks)

- (b)** Show that the refractive index of glass with a critical angle of 45° is 1.41.

(2 marks)

7 (a) Fig. 6.1 represents a sound wave of wavelength 0.45m travelling from left to right.

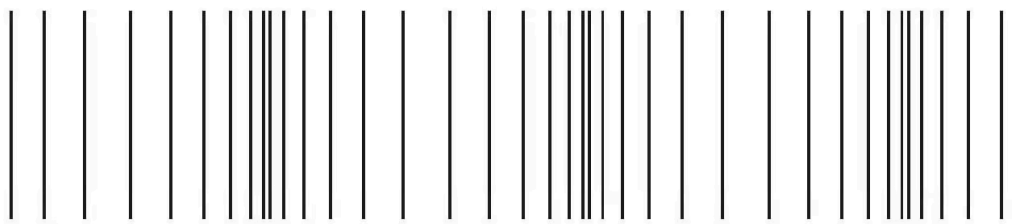


Fig. 6.1 (not to scale)

On Fig. 6.1:

(i) at the centre of a compression, mark a cross and label it C.

[1]

(ii) at the centre of a rarefaction, mark a cross and label it R.

[1]

(iii) draw a double-headed arrow to represent a distance of 0.90 m.

[1]

(3 marks)

- (b)** The frequency of the wave is 750 Hz.

Calculate the speed of the wave.

speed =

.....
.....
(2 marks)

- (c)** Suggest a medium through which the sound wave is travelling and state your reasoning.

.....
(1 mark)

- (d)** Another type of wave that consists of compressions and rarefactions is ultrasound.

(i) State one other similarity between sound of frequency 750 Hz and ultrasound.

[1]

(ii) State one way in which sound of frequency 750 Hz is different from ultrasound.

[1]

.....
.....
(2 marks)

8 (a) A teacher demonstrates the action of a device. Fig. 10.1 shows the symbol for the device.

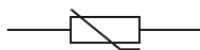


Fig. 10.1

State the name of this device.

(1 mark)

(b) Fig. 10.2 shows another device being used in a circuit. The circuit contains a 6.0 V lamp.

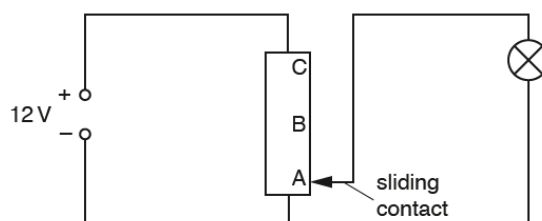


Fig. 10.2

(i) The sliding contact of this device is at position **A**, as shown in Fig. 10.2.

Describe and explain the brightness of the lamp when the sliding contact is in this position.

[2]

(ii) The teacher moves the sliding contact from position **A** to position **B**. Describe and explain what happens to the brightness of the lamp.

[2]

(iii) The teacher moves the sliding contact from position **B** to position **C**. Suggest what happens to the lamp.

[1]

(5 marks)

9 (a) (i) Describe what is meant by an electric field.

[1]

(ii) State what is meant by the direction of an electric field.

[1]

(2 marks)

(b) Fig. 8.1 shows a polystyrene ball covered with aluminium paint. The polystyrene ball is suspended between two charged metal plates by an insulated thread.

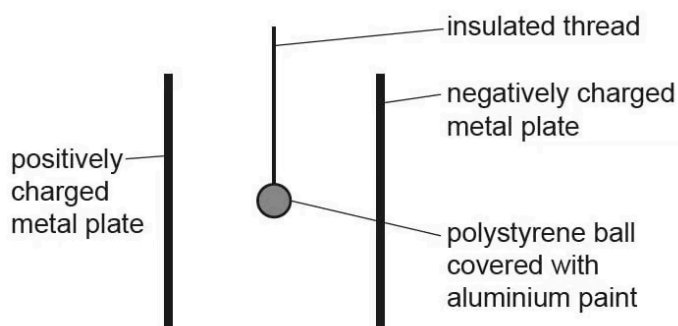


Fig. 8.1

The ball oscillates between the two charged plates.

Explain why the ball oscillates.

(4 marks)

(c) There is a current of 0.29 A in an electrical circuit.

Calculate the time taken for a charge of 15 C to flow through the electrical circuit.

time =

.....

.....

.....

(3 marks)

10 (a) The isotope hydrogen-1 has a proton number of 1 and a nucleon number of 1.

Two isotopes of helium are helium-3 and helium-4.

Helium-3 has a proton number of 2 and a nucleon number of 3.

Helium-4 has a nucleon number of 4.

Complete Table 11.1 for neutral atoms of these isotopes of helium.

Table 11.1

	helium-3	helium-4
number of neutrons		
number of electrons		
mass compared to a neutral atom of hydrogen-1		

(3 marks)

(b) An experiment takes place in a laboratory shielded from all background radiation. A sample of radioactive material is wrapped in aluminium foil of thickness 0.1 mm. A detector of ionising radiation placed 1 cm from the foil records a reading.

A piece of aluminium of thickness 5 mm is placed between the detector and the foil. The detector reading drops to zero.

State and explain any type of radiation passing through the aluminium foil.

(3 marks)

11 (a) Edwin Hubble was an American scientist who studied galactic recession.

Define the Hubble constant, H_0 .

.....
(1 mark)

(b) State the current value of the Hubble constant.

.....
(2 marks)

(c) The Triangulum galaxy is the Milky Way's second closest galactic neighbour. The Triangulum galaxy is approximately 2.7 million light years from Earth.

Calculate the speed with which the Triangulum galaxy is receding from Earth. You should give your answer in km/s.

..... km/s

.....
(4 marks)